Experiment

① 8 bit - Addition & Subtraction

A = 9C H

A = 1001 1100

B = 7A H

B = 0111 1010

Addition: A + B => (9C + 7A) H.

A + B => 1001 1100

0111 1010 (+)

10001 0110

A + B = 16 H

Cy = 1.

Subtraction: A - B => (9C - 7A) H.

A - B => 1001 1100

0111 1010

00000100

A - B = 04 H

8-bit Multiplication &

B) 16-bit Addition & Subtraction

X = 1234 H

Y = 5678 H

X+Y = 1234 + 5678

LSB = (34 + 78) H.

LSB ⇒ 1000/0

100/110

LSB => 00110100

+ 01111000

10101100

MSB = 12H + 56H.

MSB ⇒

MSB => 00010010

01010110

01101000

MSB => 68 H.

MSB + LSB ⇒ 0110100010101100

X+Y ⇒ 68 AC H.

Subtraction.

X-Y ⇒ 1234 H - 5678H

⇒ -4444 H.

y into 2's complement.

y = 01010110 01111000

1s complement ⇒ 1010 1001 1000 0111

Add 1 ⇒ 1010 1001 1000 1000

X-Y ⇒ 0001 0010 0011 0100

1101 0110 0111 1000

1011 1011 1011 1100

X-Y ⇒ BBBCH

8-bit Multiplication & Division

Multiplication : 12H x 05 H.

Hexadecimal to decimal.

12 H => (1x16¹) + (2x16⁰)

= 16 + 2.

= 18

05 => (0x16¹) + (5x16⁰)

=> 5

18 x 5 => 90

Decimal to Hexadecimal.

90 ÷ 16 => 5 (10)

=> 5A H.

Division: 64H ÷ 05 H.

64 H => (6x16¹) + (4x16⁰)

=> 96 + 4

=> 100

05 H = 5

100 ÷ 5 => 20

20 ÷ 16 => 14 H.

16-bit Multiplication & Division

~~=> 5678H * 5678H~~

=> 1234H x 5678H

1234H => $(1\times16^3)+(2\times16^2)+(3\times16^1)+(4\times16^0)$

=> 4096 + 512 + 48 + 4

=> 4660

5678H => $(5\times16^3)+(6\times16^2)+(7\times16^1)+(8\times16^0)$

=> (20480 + 1536 + 112 + 8

= 22136

1234H x 5678H => 4660 x 22136

=> 10315376O

Convert into Hexadecimal

10315376O => 626006OH.

Division.

5678H => 22136

1234H => 4660

Division => 22136 ÷ 4660

=> '4 (3496).

=> 4DA8H.

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Samples:

① A = 5AH , B = 27H.

Addition : A + B => 5AH + 27H

LSB = A + 7

= 11H.

MSB => (5+2)H & Carry 1

=> 5 + 2 + 1

= 8.

A + B = 81H.

X = 1234H , Y = 4321H.

Addition => 1234
4321
5555H

Sub : 1234
4321
CF13

② A = E8H , B = 9CH.

Addition => E8H => 11101000
9CH 10011110
11100100

A + B => 84H.

Subtraction: 1110 1000
1001 1100
0100 1100 => 4CH.

③ Case 1: A = 14H, B = 06 H

1) Multiplication

A = (1 x 16) + (4 x 16°)

= 20

B = 6 H

A x B => 20 x 6 => 120

120 ÷ 16 => 7 (8)

= 78 H.

2) Division,

14 H ÷ 06 H.

20 ÷ 6 => 3 (2)

=> 03 H, 02H.

Case 2: A = F2H, B = 0AH.

A => 242, B = 10

A x B = 2420

Hexadecimal: 2420 ÷ 16 => 0974 H

Division.

242 ÷ 10 => 24 (2)

24 ÷ 16 => 1 (8) H

2 => 02 H.

5 FBH x 07 H

FB H => 251₁₀ => 1757H

=> 06H DDH